

Consumer Complaint Urgency Classification Using Natural Language Processing

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1. Introduction

1.1 Background

Financial-service complaints contain descriptions of situations that differ substantially in urgency. Some narratives report routine enquiries or minor inconveniences, while others describe active identity theft, unauthorized transactions, loss of access to essential funds, imminent housing consequences, or other continuing harm. Complaint-handling teams therefore face more than a categorization problem: they must identify which narratives require rapid attention while still processing a large and varied volume of text.

The Consumer Financial Protection Bureau (CFPB) publishes complaint data concerning consumer financial products and services. Its Consumer Complaint Database contains structured fields and, where consumers consent to publication, narrative descriptions in the consumers' own words. The CFPB states that it takes steps to remove personal information before narratives are published. It also cautions that the database is not a statistical sample and is not necessarily representative of all consumer experiences (Consumer Financial Protection Bureau [CFPB], n.d.-a, n.d.-b). These narratives nevertheless provide a valuable public resource for studying natural language processing (NLP) methods applied to real consumer issues.

Text classification converts unstructured language into predefined labels. Traditional methods commonly represent text with term frequency–inverse document frequency (TF-IDF) and apply classifiers such as Multinomial Naive Bayes, Logistic Regression, or Support Vector Machines. Transformer models instead learn contextual representations from token sequences. DistilBERT is a compressed transformer intended to retain much of BERT's language capability while reducing model size and inference cost (Sanh et al., 2019). Comparing these approaches under one controlled experimental design can show whether contextual modeling provides enough benefit to justify its greater computational requirements for complaint urgency triage.

This project develops a three-class NLP system that predicts whether a complaint narrative has **Low**, **Medium**, or **High** urgency. Low urgency denotes a routine, resolved, informational, or minor issue without continuing material impact. Medium urgency denotes a significant unresolved problem requiring priority attention but without explicit immediate severe harm. High urgency denotes immediate or continuing financial, identity, housing, legal, safety, or account-access risk. These labels are operational research annotations created for this assignment; they are not official CFPB findings, legal conclusions, or recommendations for real-world automated decision-making.

1.2 Problem Statement

Existing complaint research has examined product classification, company-response outcomes, topic discovery, and broad complaint detection. Those tasks do not directly answer a safety-oriented operational question: **how urgently should a complaint narrative be reviewed based on the harm described in its full text?** Product categories alone cannot provide this answer because two complaints about the same product may describe very different degrees of immediacy and harm.

Consumer tone is also insufficient; a calm narrative may describe severe continuing fraud, while an angry narrative may concern a resolved inconvenience.

The technical challenge is compounded by ambiguous language, long narratives, and class imbalance. An urgency classifier that achieves high overall accuracy by favoring the majority Medium class could still perform poorly on Low or High complaints. Missing a High-urgency case is particularly important because the narrative may describe continuing harm. At the same time, overpredicting High urgency could reduce the usefulness of triage by overwhelming reviewers. The study therefore requires common data splits, class-sensitive metrics, leakage controls, and class-level error analysis rather than accuracy alone.

A further challenge is that the CFPB dataset does not contain an official urgency label. A transparent annotation process was therefore required before supervised learning could begin. The project used two independent AI annotators under a written guide, measured agreement, refined the guide through calibration, adjudicated disagreements, recorded uncertainty, and preserved label provenance. Because the labels are AI-generated, the project must disclose their origin and treat them as constructed study targets rather than objective ground truth.

1.3 Aim, Objectives, and Research Questions

The aim of this project is to design and evaluate an NLP-based complaint urgency-classification system and demonstrate the selected models in an interactive academic prototype.

The objectives are:

- To acquire and reproducibly prepare 3,000 unique English CFPB consumer complaint narratives.
- To define Low, Medium, and High urgency through a documented annotation protocol and evaluate inter-annotator agreement.
- To implement Multinomial Naive Bayes, Logistic Regression, calibrated Linear SVM, and DistilBERT using a fixed training, validation, and test split.
- To compare the four models using accuracy, macro and weighted precision, recall, and F1-score, with particular attention to High-urgency recall and High-to-Low errors.
- To compare model efficiency using training time, prediction time, approximate model size, and explanation time.
- To build a Streamlit prototype that supports individual and comparative predictions, calibrated confidence where available, response guidance, input validation, and local LIME explanations.
- To identify limitations, error patterns, and practical implications without presenting the prototype as a production decision system.
- The study addresses three research questions:
 - How effectively can complaint narratives be classified into Low, Medium, and High urgency using classical TF-IDF models and DistilBERT?
 - How do the four methods compare in macro-F1, overall performance, High-urgency recall, efficiency, and serious error patterns?

- What trade-offs between predictive performance, interpretability, confidence calibration, and computational cost are relevant to an academic urgency-triage prototype?

1.4 Significance and Contributions

The project contributes a complete, traceable study design for complaint urgency rather than reusing existing CFPB product or response labels. Its annotation guide focuses on current harm and inability to stop harm, providing an operational distinction between routine issues, significant unresolved issues, and time-sensitive severe risk. The project also records agreement and calibration instead of treating generated labels as automatically reliable.

Methodologically, all four classifiers use the same fixed complaint IDs for training, validation, and testing. The three classical methods share one training-fitted TF-IDF feature space, while DistilBERT processes the original narratives using its own tokenizer. Macro-F1 is the primary selection and comparison metric because each urgency class should contribute equally despite imbalance. High-urgency recall and High-to-Low errors add a safety-oriented perspective that overall accuracy cannot provide.

The intended practical contribution is an educational prototype that allows users to compare different NLP methods on the same text and inspect local explanations. The system is not designed to replace human reviewers. Instead, it demonstrates how predictive models, calibrated confidence, explanations, and response guidance could be combined while clearly communicating uncertainty and limitations.

2. Related Work

2.1 Consumer Complaint Data and NLP Applications

The CFPB Consumer Complaint Database is an official public source for complaints about financial products and services. Published fields include complaint dates, products, issues, companies, responses, and consumer narratives when publication consent is provided. The CFPB uses complaint data to understand marketplace problems, support supervision and enforcement, inform rules, and educate consumers (CFPB, n.d.-a, n.d.-b). However, it warns that complaint narratives reflect consumers' descriptions, are not independently verified by the Bureau, and should not be treated as a representative statistical sample. These properties make the dataset valuable for NLP research while requiring careful interpretation.

Prior work demonstrates several ways to analyze CFPB complaints. Vaishnav et al. (2024) presented a platform that used machine-learning methods to predict response timeliness and response type and used Latent Dirichlet Allocation to discover themes. This work shows that structured outcomes and narrative signals can support complaint analytics, but its prediction targets differ from the present study's urgency labels. Roumeliotis et al. (2025) examined reasoning large language models for detecting and classifying consumer complaints. Their work reflects a broader movement from conventional models toward general-purpose language models, but it does not eliminate the need for task-specific operational definitions, controlled data splits, and transparent evaluation.

The current project differs from product classification and response prediction because urgency depends on the described immediacy and severity of harm. A complaint about credit reporting, for example, could range from a routine inquiry to an unresolved reporting error or an active identity-theft event. The same product label therefore spans multiple urgency classes. This motivates narrative-level classification using the full complaint rather than a direct mapping from product or issue metadata.

2.2 Classical Text Representation and Classification

TF-IDF represents a document by weighting terms according to their frequency within the document and their rarity across the collection. Common terms receive less weight, while terms that are more distinctive to a document receive greater weight. Scikit-learn's `TfidfVectorizer` converts raw documents into a sparse TF-IDF matrix and supports word n-grams, frequency thresholds, normalization, stop-word removal, and sublinear term frequency (Scikit-learn Developers, 2026a). Sparse TF-IDF is computationally efficient and provides a strong baseline for many text-classification problems.

Multinomial Naive Bayes estimates class probabilities under a conditional-independence assumption. Although this assumption is unrealistic for natural language, the method is fast, works with non-negative sparse features, and offers a useful probabilistic baseline. Its smoothing parameter controls how unseen or rare features affect the estimates. However, it may favor majority patterns and may not model interactions between phrases adequately.

Logistic Regression is a discriminative linear classifier that estimates class probabilities from a weighted combination of features. With class weighting, it can give minority-class training examples more influence. Its coefficients can also support feature-level interpretation. In high-dimensional text spaces, regularization is important because many correlated unigram and bigram features may be present.

Linear SVM learns decision boundaries that maximize the margin between classes. Linear margin-based classifiers are well suited to high-dimensional sparse text representations, but a standard Linear SVM produces decision scores rather than probabilities. When an interface displays confidence, these raw scores should not be presented as probabilities. The project therefore wraps Linear SVM in sigmoid calibration using five-fold cross-validation on training data. Calibration maps decision scores to probability-like outputs while maintaining separation between fitting and held-out evaluation.

These three algorithms form a meaningful classical comparison: Naive Bayes supplies a lightweight generative baseline, Logistic Regression supplies a regularized probabilistic linear model, and Linear SVM supplies a margin-based model with explicit calibration. Because they use the same normalized text and TF-IDF matrix, performance differences are attributable mainly to the classifiers rather than different feature representations.

2.3 Transformer Models and DistilBERT

Transformers use self-attention to produce contextual token representations. Unlike bag-of-words methods, they can distinguish a term's meaning according to surrounding language and word order.

DistilBERT was produced through knowledge distillation during pretraining. Sanh et al. (2019) reported a model that was smaller and faster than BERT while retaining much of its language-understanding performance. Hugging Face provides a DistilBertForSequenceClassification architecture with a sequence-classification head and supports fine-tuning with class labels (Hugging Face, 2026).

Context is relevant to urgency because phrases such as “cannot access,” “was resolved,” “still being charged,” or “deadline tomorrow” change the interpretation of nouns such as account, payment, or report. DistilBERT may capture such relations more effectively than independent TF-IDF features. Its disadvantages include greater training cost, more complex inference, sensitivity to optimization choices, and a maximum input length. Long complaint narratives therefore require a documented truncation or chunking policy.

This project uses overlapping chunks rather than silently discarding all text beyond the first segment. Each narrative is tokenized into chunks of up to 256 tokens with a stride of 64 tokens. The model produces logits for each chunk, and the document prediction is based on mean-pooled chunk logits. This strategy retains more of long narratives but increases computation and assumes that averaging is an appropriate aggregation method.

2.4 Agreement, Evaluation, Calibration, and Explanation

Supervised learning depends on label quality. Percentage agreement alone does not account for agreement expected by chance, particularly when one class is common. Cohen's kappa measures agreement between two annotators beyond chance (Cohen, 1960). Kappa does not guarantee label validity, but a change in kappa after revising an annotation guide can provide evidence that decision rules became more consistently applicable.

For imbalanced multiclass classification, accuracy may obscure poor performance on minority classes. Macro-averaged metrics calculate a metric separately for each class and then take an unweighted mean, whereas weighted averages weight class results by support. Macro-F1 is therefore appropriate as the main metric for this study because Low, Medium, and High urgency each affect the score equally. Weighted-F1 and accuracy remain useful descriptions of overall behavior, but they are interpreted together with per-class precision, recall, and F1.

Local Interpretable Model-agnostic Explanations (LIME) approximate a model around one prediction using perturbed samples and a simpler interpretable model (Ribeiro et al., 2016). LIME can highlight words that locally support or oppose a predicted class, but it does not identify causal reasons and can vary with the perturbation sample. The prototype therefore labels explanations as local approximations, and the final report considers explanation time and stability as limitations.

2.5 Research Gap and Justification

The reviewed literature establishes that CFPB narratives can support text classification, response prediction, topic discovery, and newer language-model approaches. However, the reviewed studies do not provide the exact combination used here: a three-level urgency target based on immediate and continuing harm; a traceable AI-assisted labeling protocol with measured agreement and

adjudication; four models evaluated on identical fixed splits; validation-only configuration selection; safety-oriented High-urgency measures; and an interactive comparison prototype with calibrated confidence and local explanations.

The study is justified because urgency is distinct from product category, sentiment, topic, and response outcome. A controlled comparison can show whether a contextual transformer improves minority-class performance enough to justify its cost and can expose cases where seemingly strong accuracy hides weak Low- or High-class behavior. The study also demonstrates responsible limitations: AI-generated labels are disclosed, test data remain sealed during model selection, and the prototype is framed as academic decision support rather than an autonomous financial system.

3. Methodology

3.1 Research Design and System Flow

The project follows a supervised multiclass text-classification design. Work proceeds from official-source acquisition through cleaning, annotation, splitting, preprocessing, model selection, final evaluation, error analysis, and prototype demonstration. The train split is used to fit representations and models; the validation split is used to select configurations; and the test split is reserved for one final evaluation after configurations are frozen.

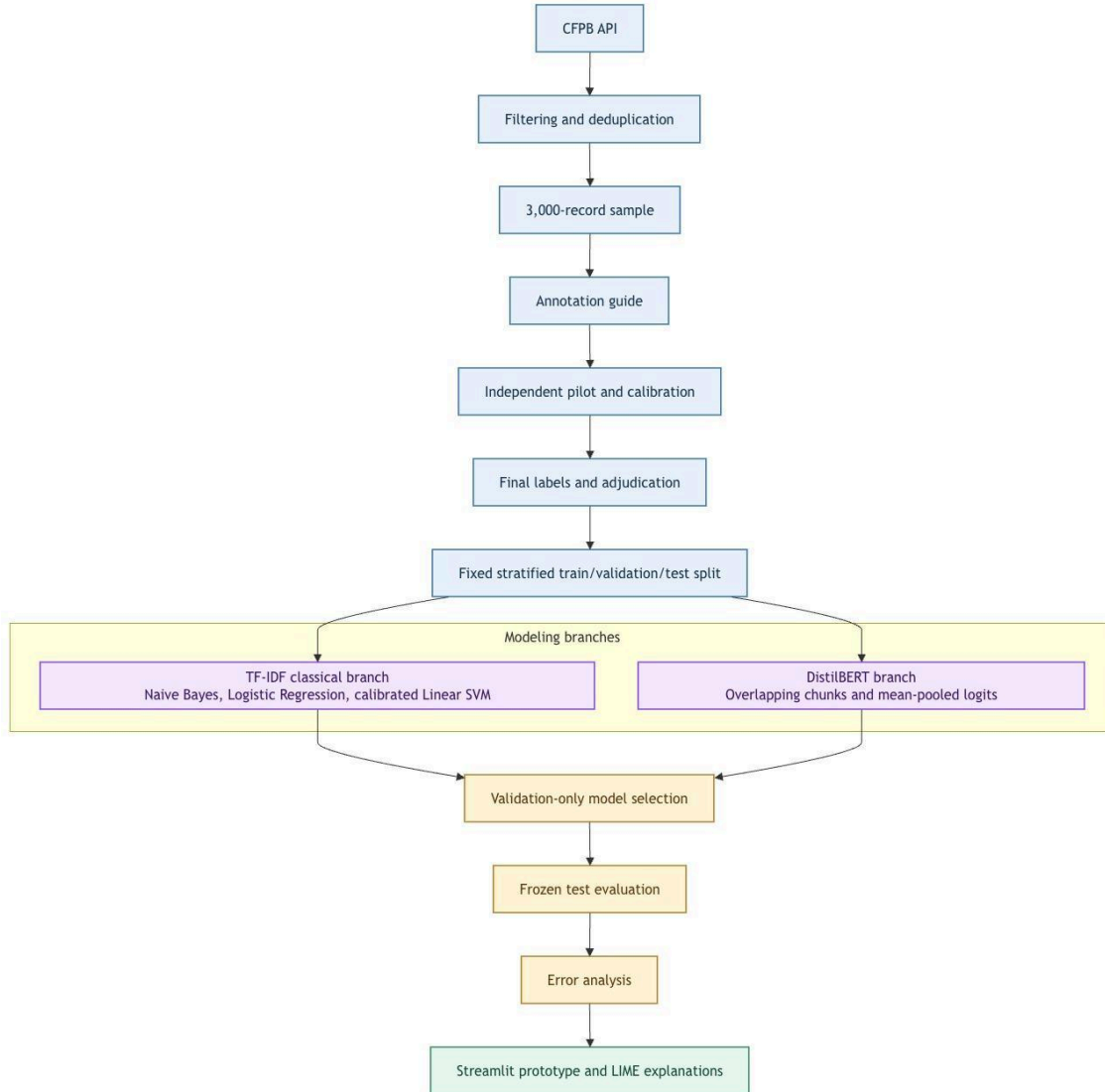


Figure 1. System workflow for CFPB complaint urgency classification.

3.2 Dataset Source and Acquisition

Data were acquired from the official CFPB Consumer Complaint Database API. The reproducible acquisition program requested complaints received from 1 January 2022 through 31 December 2025 and required a published consumer narrative. Records were sorted by creation date in descending order and retrieved in pages of 100. The acquisition run occurred on 6 August 2026 and used seed 20260806 for final sampling.

The pipeline retrieved 6,000 raw records. It discarded 1,445 duplicate narratives, leaving a cleaned pool of 4,555 records. A seeded sample of 3,000 unique narratives was then created. The raw and processed files were hashed with SHA-256 and recorded with the API query, retrieval timestamp, row counts, and cleaning losses in `data/processed/cfpb_data_provenance.json`. Existing output files are protected against accidental overwrite unless an explicit overwrite option is supplied.

The dataset contains complaint ID, narrative, product information, and other retained CFPB fields. The modeling target, `urgency_label`, was not taken from CFPB; it was created by the annotation process described below. CFPB narratives are consumer-submitted descriptions and are not verified as fact by the Bureau. The data also exclude consumers who did not complain and narratives that were not published, so the dataset cannot be interpreted as representative of all financial harm.

3.3 Urgency Annotation Protocol

The team developed version 2 of an urgency annotation guide. It instructs annotators to read the full narrative, focus on present risk rather than tone or product category, choose the highest supported urgency when multiple conditions occur, and avoid inventing missing facts. Table 1 summarizes the operational definitions.

Table 1. Operational urgency-label definitions

Label	Operational definition	Illustrative indicators
Low	Routine, resolved, informational, or minor issue without ongoing material impact.	Completed inquiry, minor inconvenience, dissatisfaction without unresolved harm.
Medium	Significant unresolved issue needing priority attention without explicit immediate severe harm.	Unresolved reporting error, disputed collection, delayed investigation, improper fees, accumulating interest.
High	Immediate or continuing material financial, identity, housing, legal, safety, or account-access risk.	Active identity theft, ongoing unauthorized transactions, inability to access essential funds, imminent foreclosure or legal deadline.

Two independent AI annotators, both recorded as gpt-5.6-terra with low reasoning effort, generated the labels. **The final labels are AI-generated research annotations.** This AI provenance is a central limitation and is disclosed throughout the coursework records. Both annotators independently labeled the same 400-record pilot without seeing each other's decisions. The pilot produced agreement on 333 records and Cohen's kappa of 0.4278. The team then refined decision rules, particularly for unauthorized inquiries, unresolved credit-report items, current fraud, account access, and imminent harm.

A second independent calibration round used 67 records. The annotators agreed on 56 records and achieved kappa of 0.7485. The higher kappa provided evidence that the refined rules improved agreement beyond chance, although it does not establish expert validity. The remaining 2,600 records were divided between the annotators according to the assignment workflow. Borderline cases could be marked uncertain with a confidence score and a short evidence-based rationale. Six final records were marked uncertain and reviewed. Disagreements and uncertain cases were resolved in the annotation decision log, and every final row was validated to contain exactly one of Low, Medium, or High.

Table 2. Annotation agreement and review evidence

Stage	Records	Agreements	Raw agreement	Cohen's kappa	Purpose
Initial pilot	400	333	83.25%	0.4278	Identify ambiguity in version 1 rules.

Calibration round	67	56	83.58%	0.7485	Verify consistency after guide refinement.
Final uncertainty review	3,000 total	Not applicable	Not applicable	Not applicable	Review six uncertain cases and freeze one label per record.

The final labeled dataset and decision log include a run identifier and source hash. The labels should be interpreted as internally defined research annotations rather than official urgency judgments. A future study should compare these labels with independent human subject-matter experts.

3.4 Exploratory Data Analysis

Table 3 shows a marked class imbalance. Medium accounts for 2,148 records (71.60%), High for 562 (18.73%), and Low for 290 (9.67%). Records were not removed merely to balance the classes. Instead, the imbalance was addressed through class-sensitive metrics and, where appropriate, class weights derived from training data.

Table 3. Final urgency-class distribution

Urgency	Count	Percentage
Low	290	9.67%
Medium	2,148	71.60%
High	562	18.73%
Total	3,000	100.00%

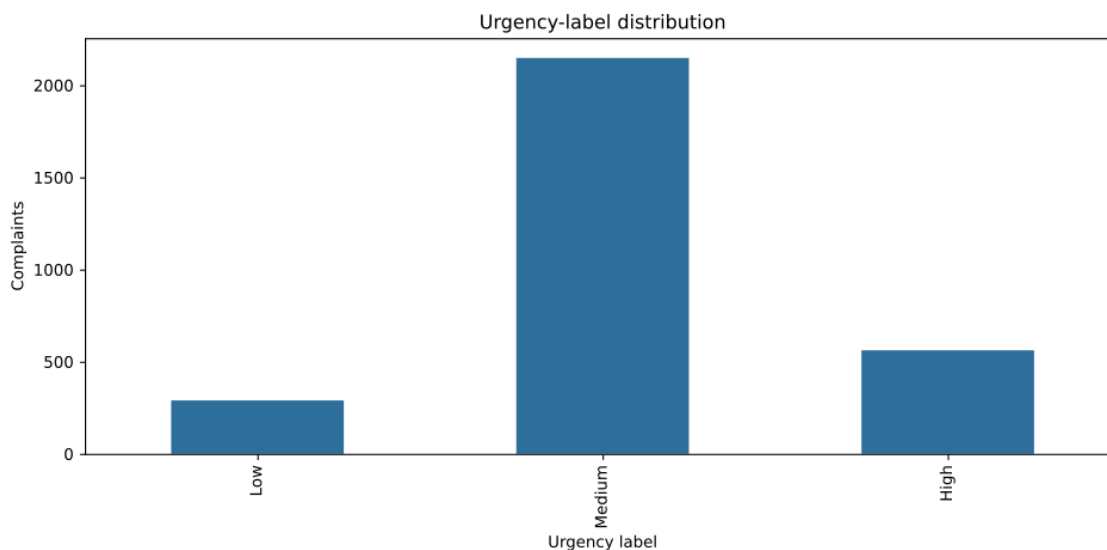


Figure 2. Distribution of Low-, Medium-, and High-urgency labels in the complete dataset. The Medium class dominates the dataset, so accuracy and weighted averages may overstate performance on the Low and High classes.

The dataset spans 11 CFPB product categories. Credit reporting or other personal consumer reports is the largest category with 1,079 complaints, followed by debt collection with 531, checking or savings

accounts with 393, and credit cards with 384. Product distribution was inspected for context, but product fields were not substituted for narrative evidence in annotation or used as the primary classical feature space.

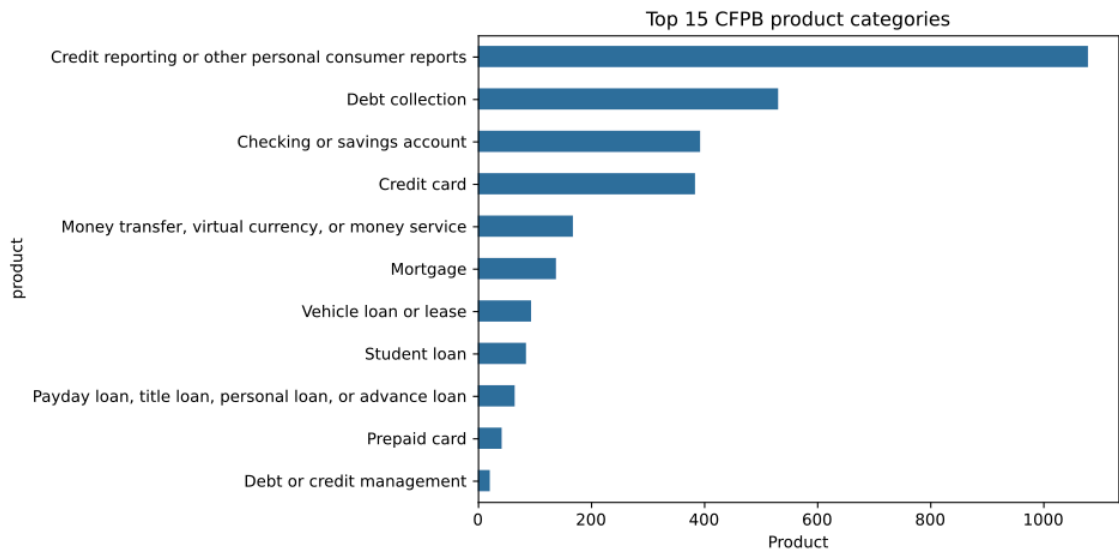


Figure 3. Distribution of CFPB product categories in the 3,000-record dataset. The concentration in credit reporting and debt collection may shape the learned vocabulary and limits confidence in generalization to less frequent product categories.

Narratives contain a mean of 207.7 whitespace-delimited tokens and a median of 159. The 95th percentile is approximately 535 tokens, and the maximum is 2,256. Character counts range from 50 to 13,141. At a 256-token limit, 43.1% of training narratives exceed the maximum sequence length. This finding motivated overlapping chunking for DistilBERT instead of using only the first 256 tokens.

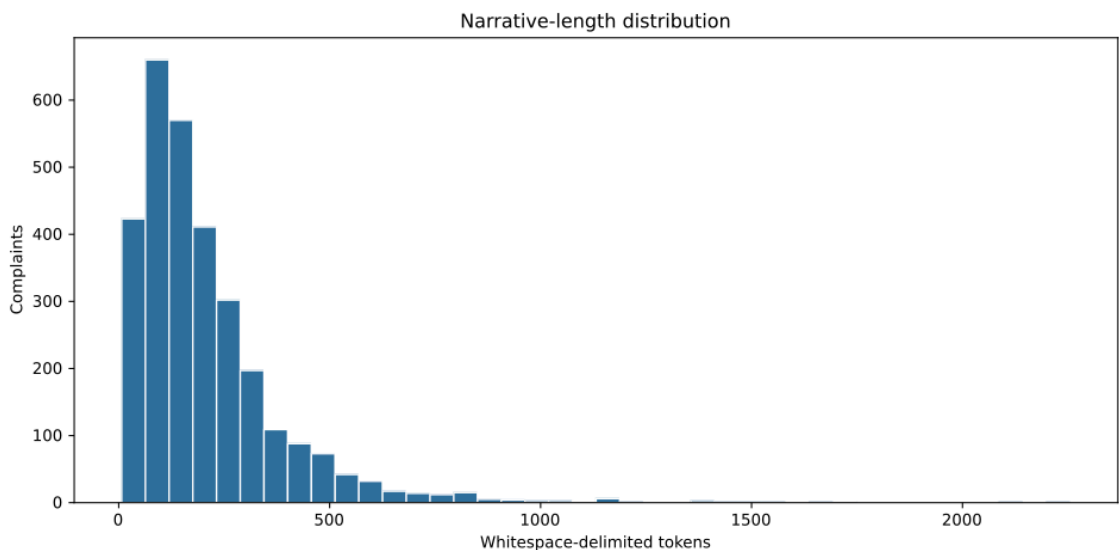


Figure 4. Distribution of complaint narrative lengths measured in whitespace-delimited tokens. The right-skewed distribution and substantial number of narratives above 256 tokens justify the documented overlapping-chunk strategy.

3.5 Fixed Data Split and Leakage Controls

After annotation and duplicate removal, the project created one stratified 70/15/15 split using seed 20260815. Complaint IDs for each subset were saved so every model uses exactly the same records. The ID sets are disjoint and their union equals the complete labeled dataset.

Table 4. Fixed stratified split by urgency class

Split	Low	Medium	High	Total	Purpose
Training	203	1,504	393	2,100	Fit vectorizer, class weights, model parameters, and calibration folds.
Validation	44	322	84	450	Select hyperparameters and DistilBERT checkpoint using macro-F1.
Test	43	322	85	450	One final evaluation after configurations are frozen.
Total	290	2,148	562	3,000	—

The TF-IDF vocabulary and inverse-document-frequency weights were fitted only on training narratives. Validation and test narratives were transformed without refitting. Model hyperparameters were selected only from validation performance. Linear SVM calibration used folds within training data. DistilBERT class weights were computed from training labels, and its checkpoint was selected by validation macro-F1. The Phase 5 report explicitly records `test_set_accessed_for_modeling: false`.

Although the Phase 4 notebook created `X_test` to verify transformation compatibility, no Phase 5 fit, prediction, metric, or selection operation used `X_test` or test labels. Final test evaluation is deferred to Phase 6.

3.6 Text Preprocessing and Representation

The classical pipeline normalizes narratives to lowercase; replaces email addresses, URLs, and numbers with placeholders; retains alphabetic tokens; collapses whitespace; and applies English stop-word removal. A word-level TF-IDF vectorizer uses unigrams and bigrams, sublinear term frequency, and `min_df=2`. It produced 27,493 features from the 2,100 training narratives.

DistilBERT uses the `distilbert-base-uncased` tokenizer and minimally altered original narratives rather than the classical normalization pipeline. Documents are divided into overlapping segments with a maximum length of 256 tokens and a stride of 64. Padding is applied to the maximum length. Chunk logits are averaged to obtain one document-level logit vector. This approach exposes more of each

long narrative to the transformer, while the overlap reduces the chance that a relevant phrase is split at a boundary.

Table 5. Text preprocessing and representation settings

Component	Classical models	DistilBERT
Input	Complaint narrative	Complaint narrative
Normalization	Lowercase; email, URL, and number placeholders; alphabetic tokens	Pretrained uncased tokenizer
Representation	TF-IDF sparse matrix	Contextual token representations
Vocabulary/features	Training-fitted 27,493 unigram/bigram features	distilbert-base-uncased vocabulary
Long documents	Full normalized text represented sparsely	256-token overlapping chunks, stride 64
Document aggregation	Not applicable	Mean of chunk logits
Leakage control	Vectorizer fitted on training only	Class weights fitted on training; checkpoint selected on validation

3.7 Algorithms and Model Selection

Four classifiers were implemented. Each model predicts one of the three urgency labels.

Table 6. Selected algorithms and their roles

Model	Main characteristic	Reason for inclusion
Multinomial Naive Bayes	Fast generative classifier for non-negative sparse features	Establishes a lightweight probabilistic baseline.
Logistic Regression	Regularized discriminative linear classifier	Provides a strong and interpretable classical baseline.
Calibrated Linear SVM	Margin-based linear classifier with sigmoid calibration	Suits sparse text while providing confidence-compatible outputs.
DistilBERT	Fine-tuned contextual transformer	Tests whether contextual representations improve urgency classification.

Classical candidate configurations were trained on X_{train} and ranked solely by validation macro-F1. The first candidate won deterministic ties. Multinomial Naive Bayes tested alpha values 0.1, 0.5, and 1.0. Logistic Regression and Linear SVM tested C values 0.1, 1.0, and 10.0. Logistic Regression used balanced class weights and a maximum of 5,000 iterations. Linear SVM used balanced class weights and five-fold sigmoid calibration.

DistilBERT was fine-tuned with inverse-frequency class weights calculated from the training labels, AdamW with learning rate 2×10^{-5} , batch size 8, and a maximum of five epochs. Validation macro-F1 was computed after every epoch. The procedure retained the best checkpoint and would stop after two consecutive non-improving epochs. The best score occurred at epoch 5, so the maximum epoch limit was reached before the patience condition triggered.

Table 7. Selected Phase 5 hyperparameters

Model	Selected configuration	Validation-selection rule
Multinomial Naive Bayes	alpha=0.1	Highest validation macro-F1 among 0.1, 0.5, 1.0.

Logistic Regression	C=1.0, balanced class weights, max_iter=5000	Highest validation macro-F1 among C = 0.1, 1.0, 10.0.
Calibrated Linear SVM	C=1.0, balanced class weights, five-fold sigmoid calibration	Highest validation macro-F1 among C = 0.1, 1.0, 10.0.
DistilBERT	LR 2e-5, batch 8, max 5 epochs, patience 2, max length 256, stride 64, inverse-frequency class weights, mean-logit aggregation	Best epoch by validation macro-F1.

The notebook emitted convergence warnings for the Linear SVM candidates. These warnings must be reviewed before final evaluation because they indicate that the optimization procedure may not have fully converged under the current iteration limit. The selected frozen configuration should not be silently retuned after viewing test performance; any convergence correction should be resolved and revalidated before the Phase 6 test-set boundary is crossed.

3.8 Evaluation Metrics

The assignment requires accuracy, precision, recall, and F1-score. This study reports macro and weighted averages, per-class metrics, and confusion matrices for all four models.

Table 8. Evaluation metrics and interpretation

Metric	Definition	Role in this study
Accuracy	Correct predictions divided by all predictions.	Describes overall correctness but is sensitive to the majority class.
Precision	True positives divided by all predictions of a class.	Measures how often a predicted urgency label is correct.
Recall	True positives divided by all actual examples of a class.	Measures how many examples of each urgency class are found.
F1-score	Harmonic mean of precision and recall.	Balances false positives and false negatives for a class.
Macro-F1	Unweighted mean of the three class F1-scores.	Primary comparison and selection metric because all urgency classes count equally.
Weighted-F1	Mean class F1 weighted by class support.	Summarizes performance in proportion to the observed distribution.
High-urgency recall	Recall calculated for the High class.	Safety-oriented measure of missed urgent complaints.
High-to-Low errors	Count of actual High complaints predicted as Low.	Identifies the most severe direction of misclassification.

Confusion matrices show all actual-versus-predicted label combinations. The final evaluation also measured training time, mean prediction time, approximate saved-model size, and LIME explanation time. These measures support a recommendation that considers practical cost and interpretability rather than macro-F1 alone.

3.9 Prototype Design and Explanation Method

The Streamlit prototype accepted a whitespace-trimmed complaint narrative from 50 through 13,000 characters. A focused two-tab flow let the user either choose one model for a single prediction or run all four frozen models on the same text. Empty input, text below 50 characters, and text above 13,000

characters were rejected with a specific message before model resources were loaded or inference began. Results were associated with the current narrative and selected model; changing either hid stale predictions and explanations until the user explicitly ran a new analysis.

The application loaded the shared TF-IDF vectorizer and three classical estimators from their frozen joblib files and loaded the frozen DistilBERT tokenizer and classifier from its artifact directory. Streamlit resource caching reused these loaded artifacts across interactions. Classical inference reused the training normalization contract through the frozen vectorizer. DistilBERT tokenized with a maximum chunk length of 256 tokens and a stride of 64, processed overflow chunks in batches of 16, averaged chunk logits at document level, and then applied softmax.

Each result displayed the predicted urgency and all three class values alongside illustrative academic guidance: standard review within five business days for Low, priority review within two business days for Medium, and immediate escalation with same-business-day human review for High. Multinomial Naive Bayes and Logistic Regression values were labelled “Model probability — not independently calibrated”; the existing calibrated Linear SVM output was labelled “Calibrated probability”; and DistilBERT output was labelled “Softmax score — not calibrated.” Phase 7 introduced no temperature scaling, recalibration, or other new calibration procedure.

LIME explanations were generated only after the user requested one for a completed single prediction. The implementation used a fixed random seed (20,260,815), 250 perturbed samples, and at most 10 displayed terms for the predicted class. The interface stated that LIME is a local model approximation, not a causal explanation, and separately reminded users that response guidance is illustrative and human review remains responsible.

3.10 Reproducibility, Ethics, and Academic Integrity

The experiment records the source dataset hash, split IDs, random seeds, Python and library versions, machine platform, hyperparameters, validation metrics, training time, and model paths. The recorded environment used Python 3.14.3 on Apple arm64 hardware with pandas 2.3.3, scikit-learn 1.9.0, Matplotlib 3.11.1, joblib 1.5.3, Jupyter 1.1.1, Transformers 4.57.6, and PyTorch 2.13.0. DistilBERT training used Apple's Metal Performance Shaders device.

The project involves public narratives from an official source. It does not attempt to re-identify consumers, and placeholder replacement during normalization reduces exposure to email-, URL-, and number-like strings for classical models. Nevertheless, public availability does not remove ethical responsibilities. Example narratives used in the report or demonstration should be paraphrased or selected carefully to avoid reproducing sensitive personal circumstances.

AI assistance was used for annotation, planning, coding, research, editing, figure generation, and draft creation. In particular, the dataset labels were created by two AI annotators, and this report draft was prepared with AI assistance from Codex. Each team member remains responsible for verifying facts, calculations, citations, interpretations, and contribution claims and for completing the official AI Usage Disclosure and Plagiarism Statement forms honestly.

4. Results and Discussion

4.1 Annotation and Data-Preparation Results

The acquisition pipeline successfully produced 3,000 unique complaint narratives from 6,000 retrieved records after removing 1,445 duplicate narratives. The annotation workflow produced a complete three-class target with traceable provenance. The pilot kappa of 0.4278 indicated only moderate beyond-chance consistency despite 83.25% raw agreement. After refining the guide, calibration kappa increased to 0.7485 with 83.58% raw agreement. The near-constant raw agreement but substantial kappa change illustrates why percentage agreement alone is insufficient when class prevalence and chance agreement differ.

The final labels are strongly imbalanced toward Medium urgency. This imbalance affects the interpretation of modeling results. A model may predict Medium frequently and obtain respectable accuracy or weighted-F1, while failing to identify Low cases or producing insufficient High recall. Macro-F1 was therefore used for selection.

The fixed split preserved approximate class proportions and generated exactly 2,100 training, 450 validation, and 450 test records. Automated checks confirmed disjoint ID sets and training-only TF-IDF fitting. These results satisfy the data-preparation requirements needed for comparable modeling.

4.2 Preliminary Validation Results

Table 9 reports Phase 5 validation metrics. These values were used to select configurations and must not be quoted as final test performance.

Table 9. Preliminary validation performance of selected model configurations

Model	Accuracy	Macro-F1	Weighted-F1	High recall	Low recall
Multinomial Naive Bayes	0.7822	0.5290	0.7434	0.5119	0.0682
Logistic Regression	0.7600	0.6576	0.7650	0.7024	0.5227
Calibrated Linear SVM	0.7956	0.6302	0.7802	0.6071	0.2727
DistilBERT	0.8044	0.7225	0.8110	0.7619	0.6818

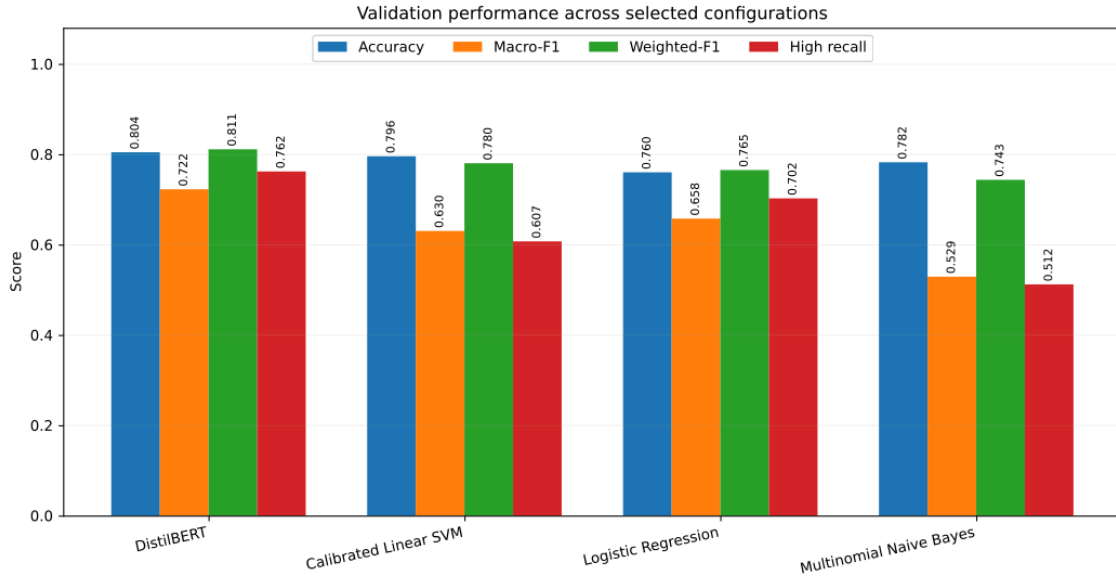


Figure 5. Validation accuracy, macro-F1, and weighted-F1 for the four selected configurations. DistilBERT led the validation summary measures, while the gap between Naive Bayes accuracy and macro-F1 illustrates how majority-class performance can conceal minority-class weakness.

DistilBERT achieved the highest validation macro-F1 of 0.7225 and the highest accuracy of 0.8044. Its class-level F1-scores were 0.5714 for Low, 0.8645 for Medium, and 0.7314 for High. It also achieved a High recall of 0.7619 and Low recall of 0.6818. These results suggest that contextual representations and long-document chunking may help distinguish minority classes. However, the conclusion is provisional because the same validation set influenced checkpoint selection.

Logistic Regression was the strongest classical model by macro-F1 at 0.6576, although its accuracy of 0.7600 was lower than the other selected models. This apparent contradiction is informative rather than problematic. Logistic Regression recovered 52.27% of Low records and 70.24% of High records, producing a more balanced class profile than Naive Bayes or the calibrated SVM. Macro-F1 rewarded this balance, whereas accuracy favored correct predictions on the dominant Medium class.

The calibrated Linear SVM achieved 0.7956 accuracy and 0.7802 weighted-F1 but only 0.6302 macro-F1. Its Medium recall was high at 0.9161, while Low recall was 0.2727 and High recall was 0.6071. The gap between weighted and macro performance indicates that the model benefited strongly from the majority class. Its convergence warnings are an additional technical concern that should be resolved before treating the selected SVM as final.

Multinomial Naive Bayes demonstrates most clearly why accuracy alone is inadequate. It achieved 0.7822 accuracy, higher than Logistic Regression, yet its macro-F1 was only 0.5290. Medium recall reached 0.9503, but Low recall was 0.0682: it identified only about 7% of actual Low validation examples. The method therefore behaved primarily as a majority-oriented baseline despite its apparently competitive accuracy.

Table 10. Per-class validation F1-scores

Model	Low F1	Medium F1	High F1	Interpretation
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Multinomial Naive Bayes	0.1200	0.8656	0.6014	Strong majority-class fit; very weak Low recognition.
Logistic Regression	0.4946	0.8333	0.6448	Most balanced classical configuration.
Calibrated Linear SVM	0.3636	0.8689	0.6581	High Medium performance but weak Low recovery.
DistilBERT	0.5714	0.8645	0.7314	Best validation balance across all three classes.

DistilBERT training improved non-monotonically. Validation macro-F1 was 0.6778 after epoch 1, fell to 0.6614 at epoch 2, increased to 0.6928 at epoch 3, fell to 0.6813 at epoch 4, and reached 0.7225 at epoch 5. The best checkpoint was therefore epoch 5. Its recorded training-loop time, including validation passes, was approximately 2,001.36 seconds (33.36 minutes) on MPS. The classical values recorded during Phase 5 were only the selected configuration's estimator .fit() calls: approximately 0.017 seconds for Naive Bayes, 0.408 seconds for Logistic Regression, and 0.494 seconds for calibrated Linear SVM. They exclude TF-IDF fitting and the full candidate search. Consequently, these values document the observed procedures but are not an apples-to-apples end-to-end training benchmark.

4.3 Final Held-Out Test Results

The four Phase 5 configurations were hashed and frozen before the test labels were opened. Each model then produced predictions for the same 450 complaint IDs. No model was retrained, tuned, recalibrated, or replaced in response to the results. Macro-F1 remained the pre-declared primary comparison metric.

Table 11. Final held-out test performance

Model	Accuracy	Macro precision	Macro recall	Macro-F1	Weighted precision	Weighted recall	Weighted-F1	High recall	High→Low errors
Multinomial Naive Bayes	0.7933	0.7779	0.5292	0.5535	0.7894	0.7933	0.7574	0.5412	0
Logistic Regression	0.7644	0.6367	0.6915	0.6589	0.7877	0.7644	0.7728	0.7647	3
Calibrated Linear SVM	0.8044	0.6624	0.5700	0.5894	0.7738	0.8044	0.7788	0.6588	1
DistilBERT	0.7733	0.6585	0.7013	0.6747	0.7920	0.7733	0.7807	0.7059	2

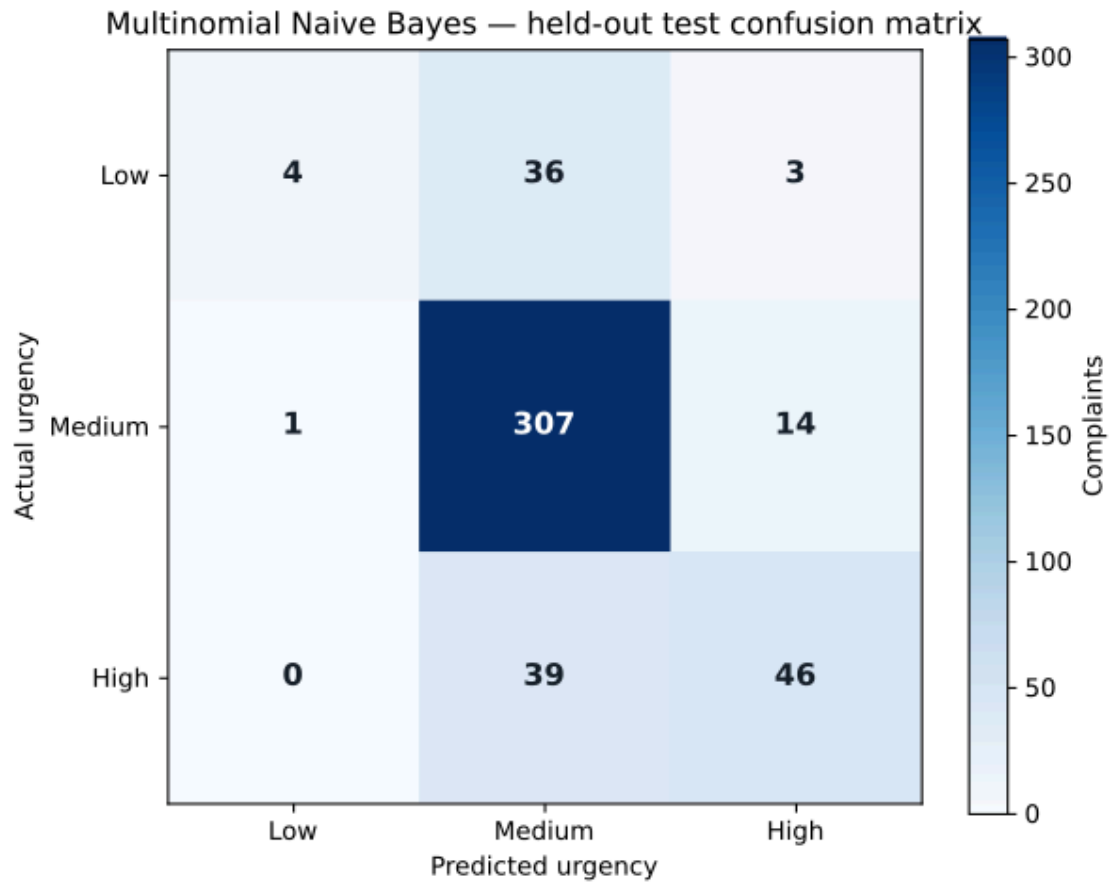


Figure 6. Confusion matrix for Multinomial Naive Bayes on the held-out test set. Naive Bayes classified 307 of 322 Medium complaints correctly but only 4 of 43 Low and 46 of 85 High complaints. It made no direct High-to-Low errors, yet routed 39 High complaints to Medium, confirming strong majority-class bias.

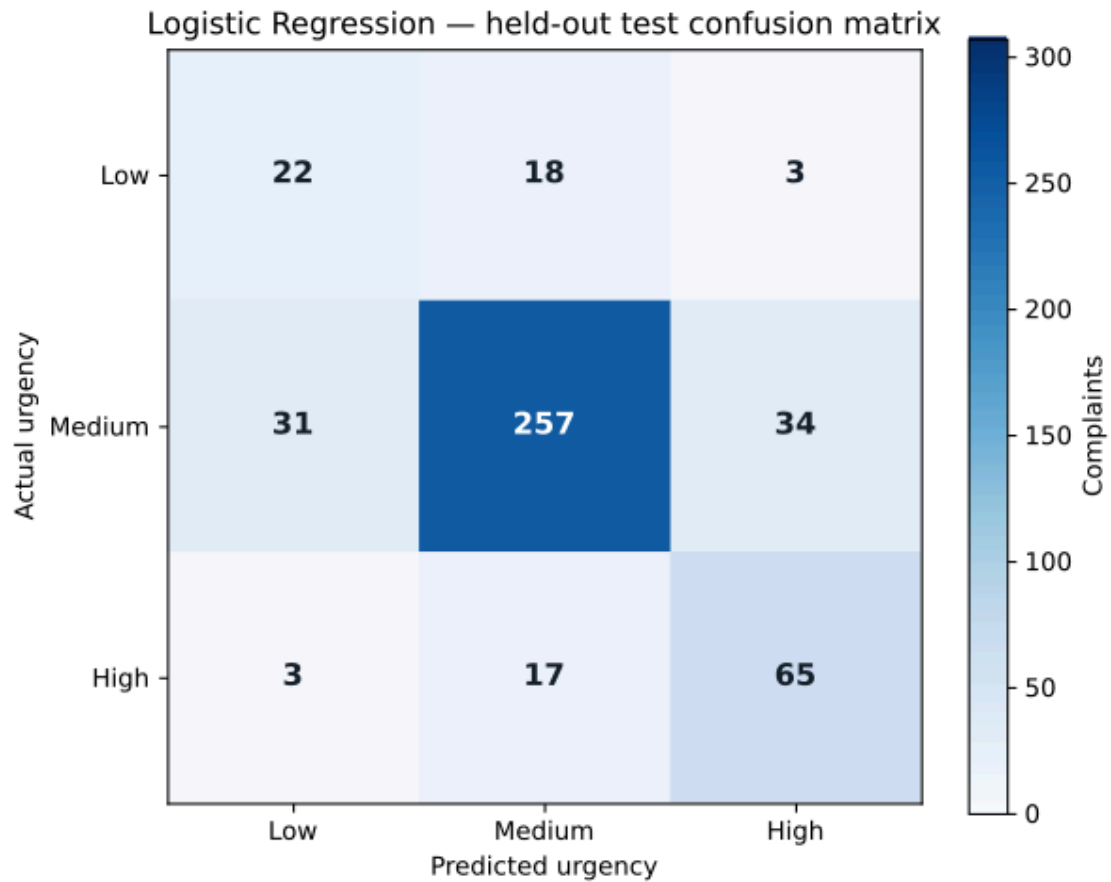


Figure 7. Confusion matrix for Logistic Regression on the held-out test set. Logistic Regression retained the most balanced classical profile: it recovered 22 of 43 Low and 65 of 85 High complaints. Its High recall was the best of all four models, although it made three direct High-to-Low errors.

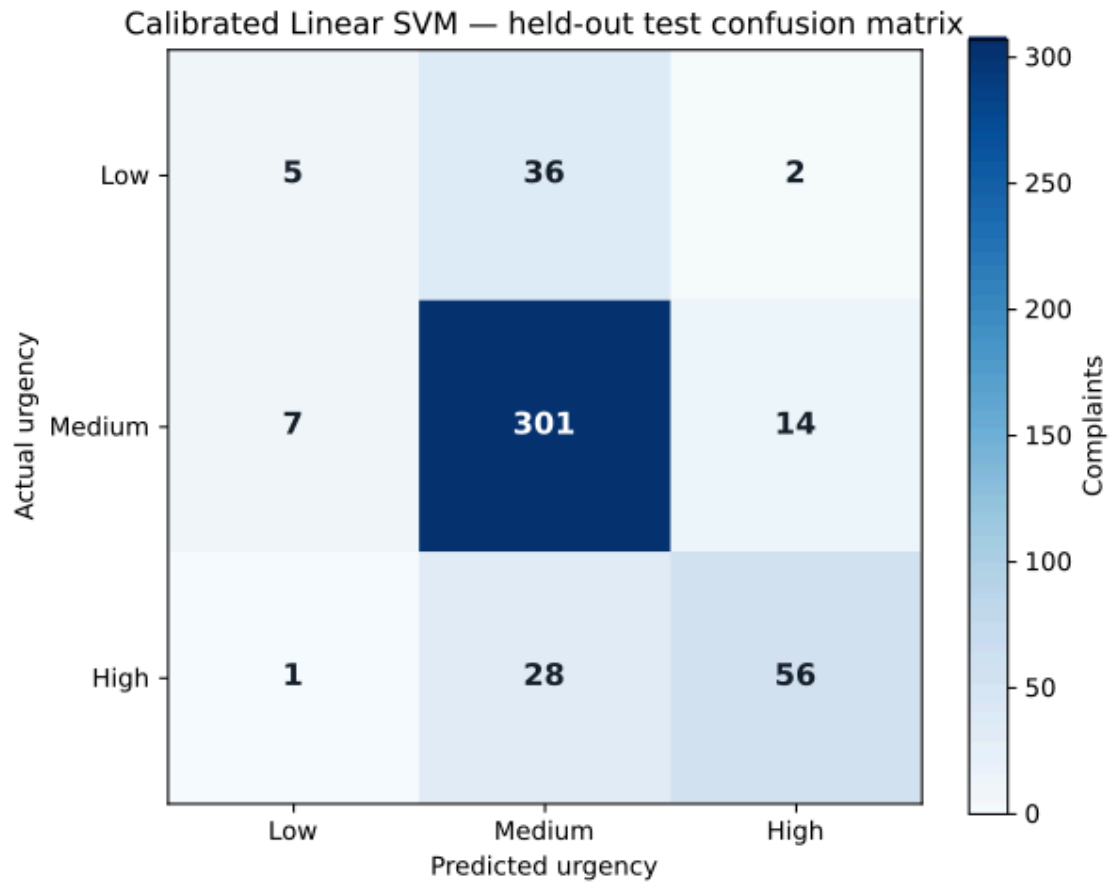


Figure 8. Confusion matrix for calibrated Linear SVM on the held-out test set. The calibrated SVM achieved the highest accuracy by correctly classifying 301 of 322 Medium complaints. However, it recovered only 5 of 43 Low complaints, producing a Low F1 of 0.1786 and a substantially lower macro-F1 than its accuracy suggests.

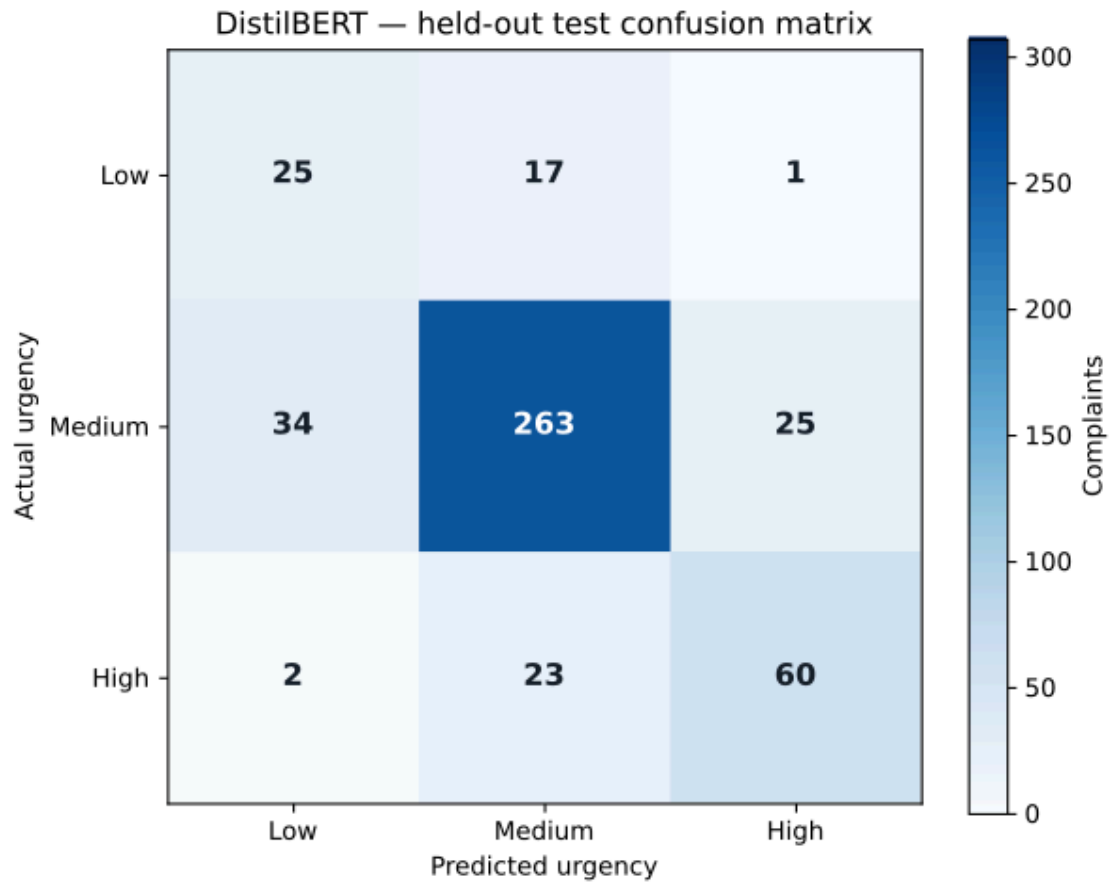


Figure 9. Confusion matrix for DistilBERT on the held-out test set. DistilBERT recovered 25 of 43 Low and 60 of 85 High complaints. It produced the best Low F1 and the highest macro-F1, although its High recall was lower than Logistic Regression and it made two High-to-Low errors.

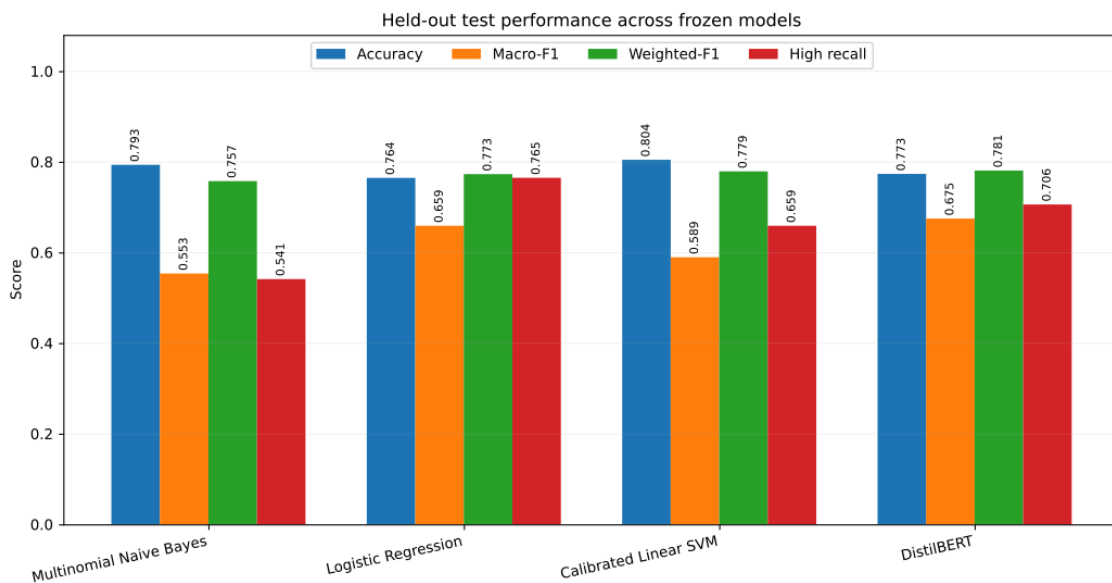


Figure 10. Final held-out test comparison of accuracy, macro-F1, weighted-F1, and High-urgency recall. DistilBERT remained the macro-F1 leader, but its test macro-F1 of 0.6747 was 0.0478 below its validation value. Logistic Regression followed at 0.6589, only 0.0158 behind DistilBERT, and achieved High recall 0.0588 higher. The SVM's leading accuracy did not translate into balanced class performance.

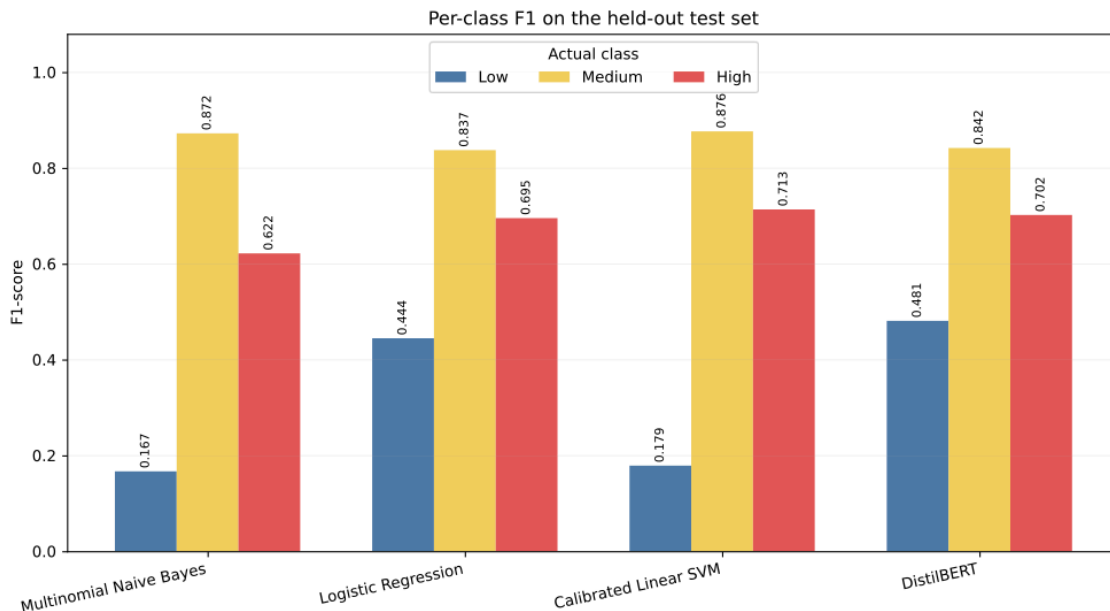


Figure 10a. Per-class F1-scores for the four frozen models. The chart makes the dominant Medium-class performance visible alongside the much weaker Low-class results. DistilBERT achieved the strongest Low F1 (0.4808), while its High F1 (0.7018) narrowly exceeded the other models.

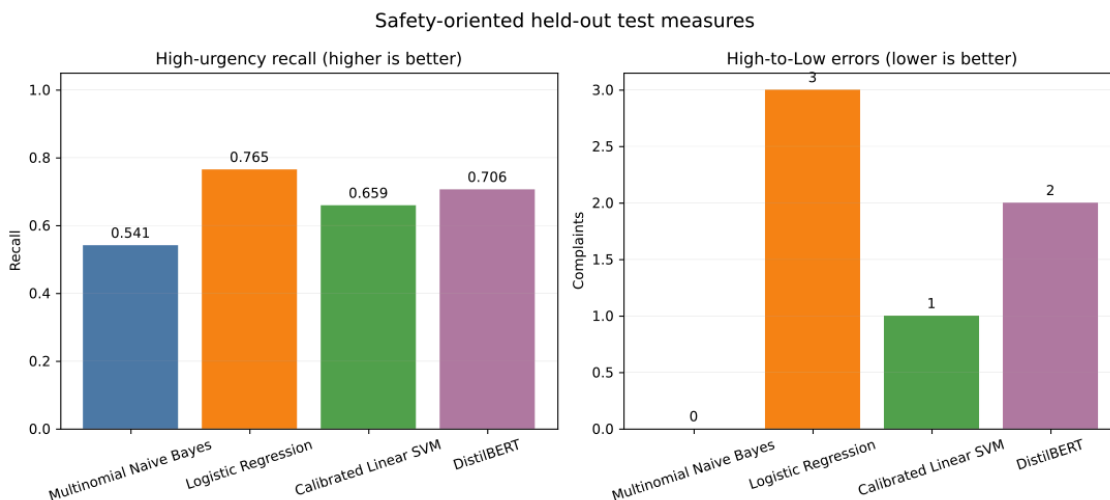


Figure 10b. High-urgency recall and direct High-to-Low errors. Logistic Regression detected the largest proportion of High complaints, whereas Naive Bayes made no direct High-to-Low errors

because most of its missed High cases were assigned to Medium. Both measures are therefore needed to interpret safety behavior.

4.4 Efficiency and Explanation Results

Artifact loading was excluded from prediction timing. After one unrecorded warm-up pass, each classical model processed the full test set ten times; its reported steady-state time is the mean total TF-IDF transformation and prediction duration divided by 450. DistilBERT processed the full test set in three timed passes after one unrecorded warm-up pass on Apple MPS with batch size 16; its timing includes tokenization, 256-token overlapping chunks with stride 64, inference over 725 chunks, and document-level mean-logit pooling. Repetition was used only to reduce timing noise; predictive metrics were calculated from one prediction set. LIME used 250 perturbations of the same correctly classified High example. The classical LIME times are means of three runs, whereas the DistilBERT value is one run because of its substantially higher inference cost. All timings are descriptive measurements from this Mac and execution session, not hardware-independent performance guarantees.

Table 12. Final efficiency and explanation comparison

Model	Recorded Phase 5 fit time (not standardized)	Mean warm-start prediction time per narrative	Approximate persisted size	Recorded LIME explanation time	Timing scope and repetitions
Multinomial Naive Bayes	0.017 s	0.204 ms	1.99 MiB	0.036 s	Fit: selected estimator only; prediction: mean of 10 runs; LIME: mean of 3 runs; size includes shared TF-IDF vectorizer.
Logistic Regression	0.408 s	0.199 ms	1.36 MiB	0.029 s	Fit: selected estimator only; prediction: mean of 10 runs; LIME: mean of 3 runs; size includes shared TF-IDF vectorizer.
Calibrated Linear SVM	0.494 s	0.208 ms	3.88 MiB	0.033 s	Fit: selected estimator only; prediction: mean of 10 runs; LIME: mean of 3 runs; size includes five calibrated fold estimators and vectorizer.
DistilBERT	2,001.36 s (33.36 min)	41.310 ms	256.34 MiB	6.714 s	Fit: five-epoch loop including validation;

					prediction: mean of 3 runs on Apple MPS (725 chunks, batch 16); LIME: 1 run.
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The fit-time column must not be used to calculate relative training-speed claims: the classical and DistilBERT measurements have different scopes. Likewise, the DistilBERT LIME value is a single observation, not a mean. The bold classical prediction values identify the smallest observed means in this session, but the small differences among classical models should not be interpreted as stable rankings without further benchmarking.

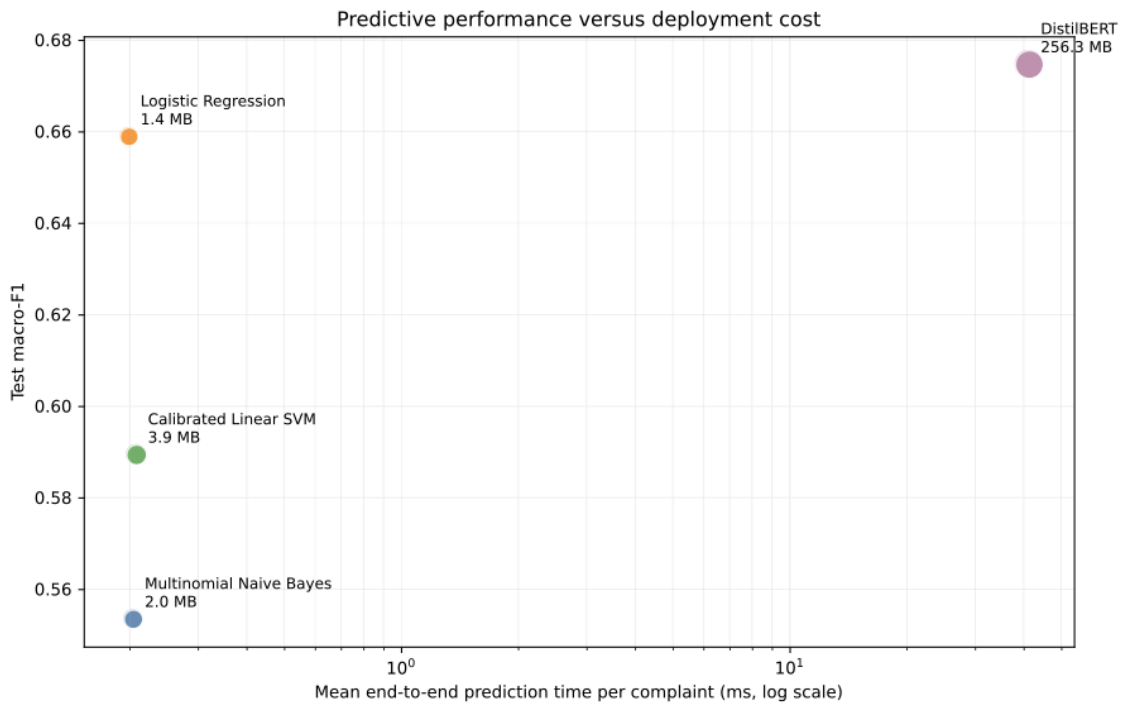


Figure 11. Predictive performance versus computational cost for the four models. DistilBERT's macro-F1 advantage over Logistic Regression was 0.0158, while its mean prediction time was approximately 207 times longer and its persisted artifact set approximately 189 times larger. The transformer remains usable for an academic demonstration, but Logistic Regression offers a substantially stronger efficiency–performance trade-off.

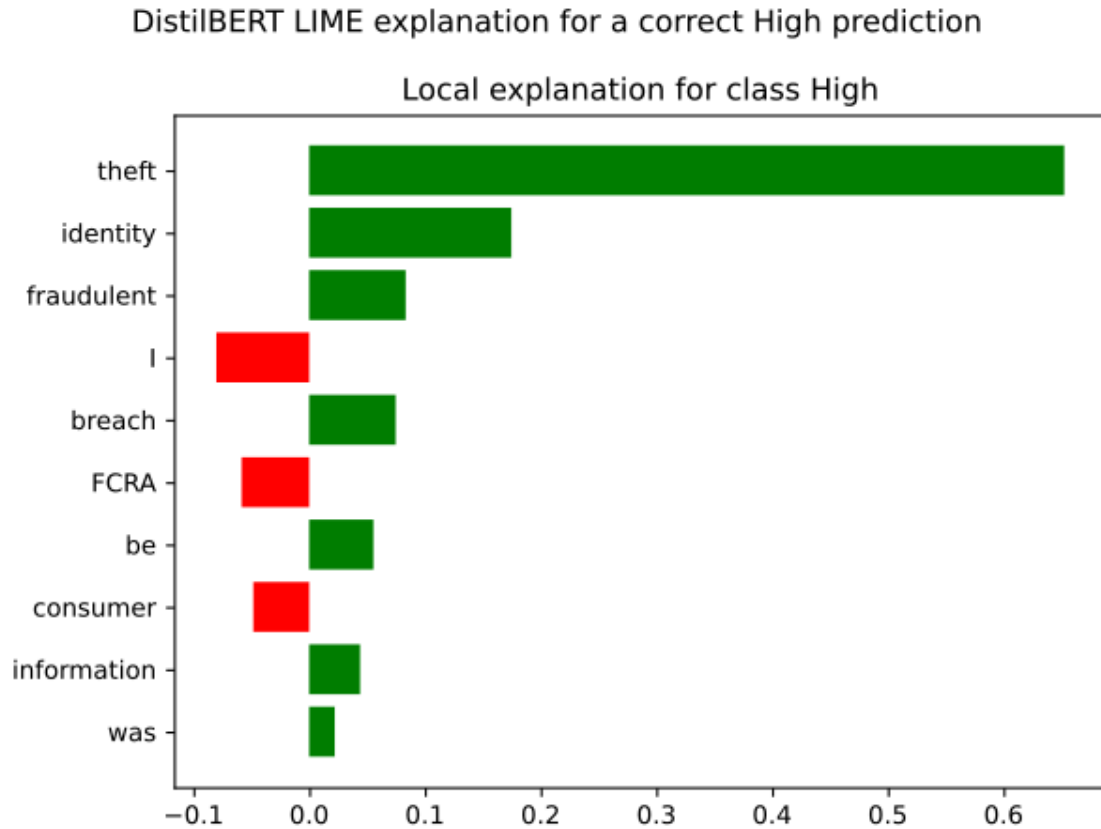


Figure 12. Representative LIME explanation for a correctly classified High-urgency complaint.

For this example, terms associated with theft, identity, and fraud locally supported the High prediction. Other terms locally opposed it. The weights describe one perturbation-based approximation around one prediction; they do not establish causal reasons, general feature importance, or factual truth.

4.5 Error Analysis

Representative errors were selected deterministically from the saved predictions after evaluation. The examples were reviewed only to understand failure patterns; they were not used to change any model. Summaries below are paraphrased and omit identifying details.

Table 13. Representative held-out test errors

Case	Actual	Predicted	Model(s)	Paraphrased evidence	Likely error factor	Implication
E1	High	Low	Logistic Regression	An unauthorized credit inquiry appeared on a consumer's report and was described as potential	Urgent harm cues may be implicit, distant, or outweighed by routine credit-report language.	A serious complaint could be under-triaged and receive an inappropriately slow response.

				identity misuse.		
E2	Low	Medium	Multinomial Naive Bayes	A familiar person-to-person transfer was temporarily held for review, and the consumer contacted the bank.	Negative event vocabulary may outweigh resolved or minor context.	Priority-review capacity may be used unnecessarily.
E3	Medium	High	Multinomial Naive Bayes	A detailed error-resolution request described an unresolved treasury-program service issue without clear immediate severe harm.	Formal and severe wording may resemble High examples despite missing immediacy.	The case may be over-escalated relative to the guide.
E4	High	Medium	Multinomial Naive Bayes	A very long mortgage narrative alleged deceptive conduct and continuing risk during a loan dispute.	Decision-critical evidence may be diluted by extensive background and mixed context.	Long narratives may hide urgent passages and require human review.

The errors separate label ambiguity from model limitations. Several narratives mix historical, resolved, and continuing events or omit timing, so adjacent labels can remain plausible even under the guide. The long-document case also shows that test-length handling is not only a transformer issue: TF-IDF can dilute sparse urgent cues within extensive context, while mean-logit chunk aggregation can similarly dilute a short urgent segment. These observations support human review for borderline and lengthy complaints.

4.6 Prototype Results

Our prototype was deployed to streamlit cloud, and can be accessed via the following URL: <https://complaint-classification-demo.streamlit.app>. The completed application launched from the repository root with `streamlit run streamlit_app.py`. The verified run opened to a populated synthetic complaint queue, made CMP-2026-0184 available as a working case, and showed a creation form that can apply the frozen DistilBERT model after valid input is submitted. The captures below use only synthetic records and a synthetic draft narrative.

Complaint operations

Daily triage workspace

Queue

My cases

Escalations

Reports

Signed in as

Alex Peterson

Triage officer

Complaint queue

Review incoming complaints and act on the next priority.

+ New complaint

Search cases

Search by case ID, issue, or product

AI priority

All priorities

Status

All statuses

CASE AND ISSUE	AI PRIORITY	STATUS	ASSIGNEE	RECEIVED AND RESPONSE DUE	
CMP-2026-0184 Unauthorized card transaction	High	New	Alex Peterson	Received 22 Aug at 17:04 UTC Due 22 Aug at 17:04 UTC	→ Open
CMP-2026-0183 Mortgage payment history is incorrect	Medium	In review	Alex Peterson	Received 22 Aug at 16:04 UTC Due 25 Aug at 16:04 UTC	→ Open
CMP-2026-0182 Debt collection account is not recognized	High	Escalated	Unassigned	Received 22 Aug at 15:04 UTC Due 22 Aug at 15:04 UTC	→ Open
CMP-2026-0181 Request for credit report dispute information	Low	New	Alex Peterson	Received 22 Aug at 14:04 UTC Due 28 Aug at 14:04 UTC	→ Open
CMP-2026-0180 Overdraft fee appears incorrect	Medium	New	Unassigned	Received 22 Aug at 13:04 UTC Due 25 Aug at 13:04 UTC	→ Open
CMP-2026-0179 Account closed without explanation	High	Escalated	Alex Peterson	Received 22 Aug at 12:04 UTC Due 22 Aug at 12:04 UTC	→ Open
CMP-2026-0177 Routine statement and address request	Low	New	Unassigned	Received 22 Aug at 10:04 UTC Due 28 Aug at 10:04 UTC	→ Open

Figure 17. Populated complaint queue. The queue presents case and issue, AI priority, status, assignee, received time, response due time, and an open action. It supports the triage workflow before any case-specific action is taken.

Complaint operations

Daily triage workspace

Queue

My cases

Escalations

Reports

Signed in as

Alex Peterson

Triage officer

Complaint queue / Case workspace

← Previous case

1 of 7

→ Next case

CMP-2026-0184

Unauthorized transactions | Debit card | Received 22 Aug at 16:44 UTC

Issue summary

Unauthorized card transaction

Source: Web form

Complaint narrative

An unauthorized card transaction remains on my account and the funds have not been returned.

Activity timeline

Complaint received

Received via Web form

22 Aug at 16:44 UTC

Add internal note

Internal note

Record a contact, decision, or next step.

Add note

AI priority

High

Priority score: 91%

Why this was flagged

Generate explanation

Weighted influencing terms

unauthorized (+0.82)

card (+0.66)

Recommended action

Escalate now and complete a same-day priority review.

Case details

Current assignee: Alex Peterson

Assign to

Assign a case owner

Response due: 23 Aug at 16:44 UTC

Status

New

Case actions

Mark in review

Escalate case

Figure 18. Case workspace for CMP-2026-0184. The selected synthetic case shows its narrative and timeline beside AI priority, Why this was flagged, recommended action, case details, and daily case actions. The workspace lets an officer investigate and record work without presenting held-out evaluation claims in the buyer-facing view.

Complaint operations

Daily triage workspace

- Queue
- My cases
- Escalations
- Reports

Signed in as
Alex Peterson
Triage officer

New complaint

Capture the complaint details and set an operational priority.

Complaint narrative

An unauthorized credit card charge remains on my account despite multiple attempts to resolve it with the bank. The amount is still missing and I need the case reviewed urgently.

Category
Billing / Transactions

Product
Credit card

Source
Web form

+ Create and triage case

Figure 19. New complaint form with synthetic narrative. The form captures the narrative, category, product, and source before Create and triage case performs automatic DistilBERT triage. The populated sample is synthetic and valid in length; it was not submitted during capture, so the displayed example does not create a persistent record.

4.7 Overall Interpretation and Model Recommendation

DistilBERT is the final performance recommendation because it achieved the highest held-out macro-F1, the pre-declared primary metric, and the strongest Low F1 while retaining competitive Medium and High results. Its macro-F1 of 0.6747 exceeded Logistic Regression by 0.0158. It also made two direct High-to-Low errors, compared with three for Logistic Regression. These results support the original hypothesis that contextual modeling and long-document chunking can improve class balance, although the improvement is modest rather than decisive.

Logistic Regression is the recommended lightweight alternative and an important comparison model for the prototype. It achieved the best High recall at 0.7647, required only 1.36 MiB including the vectorizer, and predicted approximately 207 times faster than DistilBERT under the recorded protocol. For a resource-constrained or latency-sensitive application, this trade-off could outweigh

DistilBERT's 0.0158 macro-F1 advantage. The prototype should therefore present both models without implying that one metric alone determines operational suitability.

The SVM result demonstrates why accuracy cannot determine the recommendation: its leading accuracy of 0.8044 coincided with a Low F1 of only 0.1786. Naive Bayes showed the same pattern more strongly, with 0.7933 accuracy but Low recall of 0.0930. High recall and High-to-Low counts also require joint interpretation. Naive Bayes made zero direct High-to-Low errors, but it missed 39 High complaints into Medium, so the zero count does not indicate strong safety performance.

5. Conclusion and Future Work

5.1 Achievements

The project has established a reproducible foundation for three-class complaint urgency classification. It acquired 6,000 official-source CFPB records, removed duplicate narratives, and froze a 3,000-record unique sample. It defined operational Low, Medium, and High urgency labels, used two AI annotators, recorded pilot and calibration agreement, reviewed uncertainty, disclosed AI provenance, and produced a traceable final labeled dataset.

The modeling workflow created one fixed stratified train/validation/test split, performed exploratory analysis, fitted a training-only TF-IDF representation, implemented overlapping DistilBERT chunks, and recorded the experimental environment. Four required models were implemented and selected using validation macro-F1 only. The selected configurations were then hashed and evaluated once on the same 450-record held-out test set without post-test tuning.

The final evaluation produced complete aggregate and per-class metrics, four confusion matrices, safety-oriented measures, standardized prediction and explanation timings, deployed-size estimates, and representative error analysis. DistilBERT achieved the best test macro-F1 at 0.6747; Logistic Regression was the strongest classical model at 0.6589 and achieved the best High recall at 0.7647. The results demonstrate empirically that accuracy can obscure severe minority-class weakness.

These achievements complete the analytical objectives through Phase 7 and provide the measured evidence required for the report's results and recommendation. The project also delivered a runnable Streamlit prototype with single-model and four-model flows, precise score wording, illustrative human-review guidance, on-demand LIME explanations, input validation, stale-result invalidation, automated tests, and Figures 13–16. Signed documentation forms, named contribution evidence, institutional formatting review, and final submission packaging remain outstanding.

5.2 Limitations

First, the urgency labels are AI-generated research annotations rather than expert or official CFPB judgments. Agreement and adjudication improve consistency but do not prove validity. The guide itself encodes normative decisions about what counts as immediate or severe harm.

Second, the class distribution is imbalanced, particularly for Low urgency. Class weights and macro metrics reduce but do not eliminate the effects of limited minority examples. Performance estimates for Low are based on only 44 validation and 43 test examples, so they may be unstable.

An informal post-evaluation diagnostic probe also suggests a semantic coverage gap. All deployed models predicted High for a concise synthetic narrative explicitly describing account takeover, unauthorized account access, and risks to customer assets, yet predicted Medium for a synthetic credit-report narrative that explicitly described an unauthorized and unknown loan. These probes are illustrative rather than controlled held-out evaluation results and do not alter the reported metrics. However, they suggest the training data may have taught the models to associate credit-report disputes broadly with Medium urgency without consistently learning the guide's rule that an active unauthorized account or loan is High.

Third, the CFPB database is not representative of all consumers or all financial harms. It includes people who complained, narratives approved for publication, and records handled under CFPB publication rules. Product prevalence in the sample may shape learned vocabulary, and models may not generalize to other organizations, countries, time periods, languages, or complaint channels.

Fourth, consumer narratives may omit crucial context and are not verified as accurate by the CFPB. A model can only classify the text it receives. It cannot determine legal truth, actual financial loss, or the appropriate regulatory response.

Fifth, long-document chunking uses mean-logit aggregation. This may dilute a short urgent passage among less relevant chunks or overweight repeated language. Other aggregation approaches have not yet been compared.

Sixth, model confidence is not equivalent to correctness. Although Linear SVM is calibrated, calibration quality must be measured, and DistilBERT softmax scores may remain overconfident. LIME explanations are approximate, local, potentially unstable, and non-causal.

Finally, the final estimates come from one held-out set of 450 complaints, including only 43 Low and 85 High examples. They therefore remain sensitive to sampling variation. The study does not report confidence intervals, external validation, product-wise fairness, temporal drift, or calibration-error measures. LIME timing also used one explanation repetition for DistilBERT because of its much higher cost, so explanation-time comparisons should be interpreted descriptively.

5.3 Future Work

Beyond the assignment, label quality could be strengthened through independent financial-domain or consumer-protection experts and a larger doubly annotated sample. The High and Medium subsets should be specifically audited and expanded with varied, clearly labelled examples of active unauthorized accounts, unknown loans, identity theft, and credit-report fraud. A held-out challenge set of paraphrases should test whether models apply the same High-urgency rule when the risk is described as account takeover versus as a credit-report dispute. Stratified confidence intervals or repeated external validation could quantify uncertainty more reliably than one test split. Additional models could include character n-grams for obfuscated financial language, domain-adapted

transformers, hierarchical long-document models, or cost-sensitive objectives that penalize High-to-Low errors more heavily.

Calibration should be evaluated with reliability diagrams, expected calibration error, and a held-out calibration strategy. Thresholding or abstention could route uncertain cases to human reviewers. Fairness and robustness analysis should examine performance across products, narrative lengths, time periods, and language patterns. Drift monitoring would be necessary because complaint topics and financial products change over time.

Any future deployment would require privacy review, security testing, human oversight, audit logging, organizational approval, domain-expert validation, and clear appeal or correction procedures. The prototype should remain an educational decision-support demonstration unless those requirements are met.

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